

MERIDIAN-1 DATA CENTRE, HALL A

Project No. MER-1-2026 · Issued for Construction · Revision 0 · 2026-01-15

SECTION 23 64 26 — WATER-COOLED CENTRIFUGAL CHILLERS

23 64 26 Part 1 — GENERAL

1.1 SUMMARY. Water-cooled centrifugal chillers CH-A1 and CH-A2 serving the Hall A chilled water plant, N+1.

1.2 REFERENCES. AHRI 550/590, ASME B31.9, ASHRAE 15, ISO 10816-3.

1.3 SUBMITTALS. Selection data at AHRI conditions, sound data per AHRI 575, refrigerant declaration, and weld inspection plan for chilled water piping connections.

23 64 26 Part 2 — PRODUCTS

2.1 PERFORMANCE. Coefficient of performance (COP) at AHRI 550/590 standard rating conditions shall be not less than 6.0 at 100 percent load.

2.2 CAPACITY. Each chiller shall deliver not less than 1200 TR at design conditions with zero fouling allowance deducted.

2.3 SOUND. Sound pressure level shall not exceed 85 dBA at 1 metre per AHRI 575.

2.4 REFRIGERANT. R-1233zd(E) or approved low-GWP equivalent; GWP shall not exceed 5.

2.5 PIPING WELDS. Chilled water piping welds at chiller connections: 10 percent radiographic examination per ASME B31.9; acceptance criteria zero porosity in any examined joint. Weld maps and RT reports are quality records.

2.6 VIBRATION. Peak-to-peak shaft vibration shall not exceed 4 mils at any load point, measured per ISO 10816-3 during commissioning.

23 64 26 Part 3 — EXECUTION

3.1 START-UP. Manufacturer-supervised start-up; trend logs at 25/50/75/100 percent load steps submitted to the Commissioning Authority.

3.2 COMMISSIONING. Participation in Level 3 pre-functional and Level 4 functional performance tests per Section 01 91 13.